VNR Vignana Jyothi Institute of Engineering

Electronics and Communication Engineering | Course Code: EC201

Electronic Devices & Analog Circuits - Laboratory Manual & Practical Gui

Syllabus Focus: EC201 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Semiconductor Diodes & Applications

" Syllabus Scope: Energy band theory, PN junction barrier potential, forward/reverse bias, Zener breakdow

Unit 2: Bipolar Junction Transistors BJT

" Syllabus Scope: CB, CE, CC configurations, input/output characteristics, DC load line, Q-point stabilit

EC201: Electronic Devices & Analog Circuits - Advanced Topics & Assessment

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Field Effect Transistors FET & MOSFET

" JFET pinch-off voltage, Enhancement and Depletion MOSFETs, drain and transfer characteristics, CMOS logic in

Unit 4: Small Signal Low Frequency Amplifiers

" h-parameter model of CE amplifier, voltage gain, current gain, input/output impedance, frequency response Bo

Unit 5: Operational Amplifiers & Feedback Circuits

" Ideal op-amp characteristics, inverting/non-inverting amplifiers, integrator, differentiator, instrumentatio

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: V-I Characteristics of PN Junction Diode and Zener Voltage Regulation

Lab 2: BJT Common Emitter CE Input and Output Characteristic Curves Determination

Lab 3: Frequency Response Analysis of Single-Stage BJT RC-Coupled Amplifier

Lab 4: Inverting, Non-Inverting, and Summing Operational Amplifier Circuit Verification

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Jacob Millman, Christos C. Halkias - Integrated Electronics McGraw-Hill

* Adel S. Sedra, Kenneth C. Smith - Microelectronic Circuits Oxford University Press 7th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students